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Perspective

The Curious Case of Hydrophobic Deep Eutectic Solvents: A Story on the Discovery, Design, and Applications

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ABSTRACT: Hydrophobic deep eutectic solvents (DESs) are a new generation of water immiscible solvents that have been presented in the literature for the first time in 2015. These solvents have been used for many applications. Here, an overview is given regarding hydrophobic DESs with their physicochemical properties, applications, and the challenges and limitations that the field currently is experiencing. First, a general introduction and an introduction to hydrophobic DESs are presented to explain more about DESs, their origin, and hydrophobic ones. Here, also, an overview of all the hydrophobic DESs presented in the literature is given. After the introduction, physicochemical properties such as density, viscosity, melting point, degradation temperature, volatilities, and solvatochromic properties are discussed. It is continued with the discussion of 21 different applications of hydrophobic DESs. In general, applications related to liquid−liquid extractions, liquid−liquid microextractions, formation of two and three phase systems, removal of components from leaves, gas−liquid extractions, formation of hydrogels, membrane formation, centrifugal partition chromatography, formation of a ferrofluid, coating, photoluminescence, and dye-sensitized solar cells, and catalysis have been investigated. Finally, challenges and limitations of hydrophobic DESs are discussed.



KEYWORDS: *Hydrophobic deep eutectic solvents, Densities, Viscosities, Melting points, Degradation temperatures, Volatilities, Solvatochromic properties, Applications*

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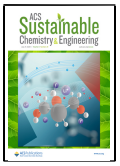
INTRODUCTION

Recently, the inceptions of sustainability, green chemistry, and engineering principles have been the pillars to achieve economic and environmental goals in both industry and academia. In simple words, research is directed towards the design of sustainable and green processes that are based on the exploitation of innovative and renewable resources, which are in balance with the least economic and environmental impacts. The design and applications of sustainable solvent media have been a hot topic in different scientific areas. Breakthrough advances in obtaining solvent alternatives to toxic and volatile organic solvents have been achieved, like the application of ionic liquids, supercritical carbon dioxide, and deep eutectic solvents (DESs), first introduced in 2003.[1] Here, we present a comprehensive overview into hydrophobic DESs, innovative solvents presented for the first time in the literature in 2015.[2,3]

INTRODUCTION TO HYDROPHOBIC DESS

DESs are eutectic mixtures that have been presented in the literature as designer solvents that could be an alternative solvent class to ionic liquids (ILs).[1] ILs have been defined as salts with the combination of an organic cation and anion with a melting point lower than 100 °C, often liquid at room temperature.[4] Finding a definition for DESs is a more daunting task. No clear and uniform definition has been reported by the major review papers on DESs, although several attempts were performed.[5−8] DESs are usually described as a combination of two or more solids that upon mixing form a liquid phase. It has been anticipated that entropy of mixing, van der Waals interactions, and hydrogen bonding plays a role in the formation of DESs. Recently, two papers were presented in the literature that investigated the phase diagrams of DESs, which correlated the phase behavior and the deepness of the eutectic point to the definition of DESs.[9,10] As expected, interactions between the components of the DES have an influence on the deepness of the eutectic point as clearly shown by the phase diagrams. Kollau et al. used regular solution theory to predict the phase diagrams and showed that some combinations of two solid components could be predicted with ideal phase behavior, while some have a clear deep eutectic.[10] It is stated that a eutectic mixture can only be considered as a DES when the melting point of the mixture is significantly lower than the one of the ideal prediction. Martins et al. state that it becomes necessary to find a stricter definition

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Table 1. Overview of All Hydrophobic Binary Deep Eutectic solvents (DESs) with Their Different References

Component 1	Component 2	Molar ratio	ref
Acetic acid	Tetrabutylammonium chloride	1:1	20
Levulinic acid	Tetrabutylammonium chloride	2:1	20, 21
Hexanoic acid	Tetrabutylammonium chloride	2:1	20
Octanoic acid	Tetrabutylammonium chloride	1:1	22
Octanoic acid	Tetrabutylammonium chloride	2:1	20, 21, 23
Decanoic acid	Tetrabutylammonium chloride	1:2	24, 25
Decanoic acid	Tetrabutylammonium chloride	1:1	22, 24
Decanoic acid	Tetrabutylammonium chloride	1.5:1	17
Decanoic acid	Tetrabutylammonium chloride	2:1	2, 20, 21, 23, 24, 17, 26, 27
Decanoic acid	Tetrabutylammonium chloride	3:1	17
Dodecanoic acid	Tetrabutylammonium chloride	2:1	22
Hexanoic acid	Tetrabutylammonium bromide	1:1	28
Octanoic acid	Tetrabutylammonium bromide	1:1	22, 28
Decanoic acid	Tetrabutylammonium bromide	1:1	22, 28
L- vs D-Glutamic acid	Tetrabutylammonium chloride	1:6	29
L- vs D-Glutamic acid	Tetrabutylphosphonium chloride	1:6	29
L-Proline	Tetrabutylammonium chloride	1:4	29
L-Arginine	Tetrabutylammonium chloride	1:8	29
n-Butyl alcohol	Tetrabutylammonium bromide	1:1	28
1-Octanol	Tetrabutylammonium bromide	1:1	28
1-Dodecanol	Tetrabutylammonium bromide	1:1	28
Oleyl alcohol	Tetrabutylammonium bromide	1:1	28
Hexanoic acid	Methyltrioctylammonium chloride	1:1	28
Hexanoic acid	Methyltrioctylammonium chloride	2:1	30
Octanoic acid	Methyltrioctylammonium chloride	1:1	28
Octanoic acid	Methyltrioctylammonium chloride	2:1	23, 30
Decanoic acid	Methyltrioctylammonium chloride	1:2	24, 25
Decanoic acid	Methyltrioctylammonium chloride	1:1	24, 28, 31
Decanoic acid	Methyltrioctylammonium chloride	2:1	2, 23, 24, 30, 31
Decanoic acid	Methyltrioctylammonium chloride	3:1	31
Decanoic acid	Methyltrioctylammonium chloride	4:1	31
Decanoic acid	Methyltrioctylammonium chloride	5:1	31
Dodecanoic acid	Methyltrioctylammonium chloride	2:1	30
Myristic acid	Methyltrioctylammonium chloride	1:1	30
Palmitic acid	Methyltrioctylammonium chloride	1:1	30
cis-9-Octadecenoic acid	Methyltrioctylammonium chloride	2:1	30
Ricinoleic acid	Methyltrioctylammonium chloride	2:1	30
1-Propanol	Methyltrioctylammonium chloride	2:1	30
1-Butanol	Methyltrioctylammonium chloride	2:1	30
Hexyl alcohol	Methyltrioctylammonium chloride	2:1	30
Capryl alcohol	Methyltrioctylammonium chloride	2:1	30
Decyl alcohol	Methyltrioctylammonium chloride	2:1	30
Dodecyl alcohol	Methyltrioctylammonium chloride	2:1	30
1-Tetradecanol	Methyltrioctylammonium chloride	1:1	30
1-Hexadecanol	Methyltrioctylammonium chloride	2:1	30
Cyclohexanol	Methyltrioctylammonium chloride	2:1	30
DL-Menthol	Methyltrioctylammonium chloride	2:1	30
n-Butyl alcohol	Methylytrioctylammonium chloride	1:1	28
1-Octanol	Methylytrioctylammonium chloride	1:1	28
1-Dodecanol	Methylytrioctylammonium chloride	1:1	28
Oleyl alcohol	Methylytrioctylammonium chloride	1:1	28
Ethylene glycol	Methyltrioctylammonium chloride	2:1	15
1-Propanol	Methyltrioctylammonium chloride	2:1	15
1,3-Propanediol	Methyltrioctylammonium chloride	2:1	15
Glycerol	Methyltrioctylammonium chloride	2:1	15
1-Butanol	Methyltrioctylammonium chloride	2:1	15
1,2-Butanediol	Methyltrioctylammonium chloride	2:1	15
Hexyl alcohol	Methyltrioctylammonium chloride	2:1	15
Capryl alcohol	Methyltrioctylammonium chloride	2:1	15
Decyl alcohol	Methyltrioctylammonium chloride	2:1	15

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Table 1. continued

Component 1	Component 2	Molar ratio	ref
Dodecyl alcohol	Methyltrioctylammonium chloride	2:1	15
1-Tetradecanol	Methyltrioctylammonium chloride	2:1	15
Cyclohexanol	Methyltrioctylammonium chloride	2:1	15
DL-Menthol	Methyltrioctylammonium chloride	2:1	15
1-Dodecanol	Methyltrioctylammonium chloride	0−1 mol ratio of the components	16
Octylic acid/Capryl alcohol	Methyltrioctylammonium chloride	3:2:1	25
Hydroquinone	Methyltrioctylammonium chloride	1:1	32
4-Phenylphenol	Methyltrioctylammonium chloride	1:1	32
4-Cyanophenol	Methyltrioctylammonium chloride	1:2	32
4-Cyanophenol	Methyltrioctylammonium chloride	1:1	32
4-Cyanophenol	Methyltrioctylammonium chloride	2:1	32
Decanoic acid	Methyltrioctylammonium bromide	1:1	24
Decanoic acid	Methyltrioctylammonium bromide	2:1	2, 24, 33, 34
Decanoic acid	Tetraheptylammonium chloride	2:1	2, 14
Hexanoic acid	Tetraoctylammonium bromide	2:1	35
Decanoic acid	Tetraoctylammonium chloride	2:1	2, 17, 18
Decanoic acid	Tetraoctylammonium bromide	2:1	2, 24, 35−37
Decanoic acid	Tetraoctylammonium bromide	1:2	25
Ethylene glycol	Tetraoctylammonium bromide	3:1	38
1,5-Pentanediol	Tetraoctylammonium bromide	3:1	38
Decanoic acid	Trihexyltetradecylphosphonium chloride	2:1	35
Ibuprofen	Tetrahepthylammonium chloride	3:7	14
Oleic acid	Tetrahepthylammonium chloride	1:1	14
Octanoic acid	Trihexyl (tetradecyl)phosphonium tetrafluoroborate	different molar ratios were tested	39
Decanoic acid	Trihexyl (tetradecyl)phosphonium tetrafluoroborate	different molar ratios were tested	39
Dodecanoic acid	Trihexyl (tetradecyl)phosphonium tetrafluoroborate	different molar ratios were tested	39
Decanoic acid	Lidocaine	2:1	37, 13, 40, 41
Decanoic acid	Lidocaine	3:1	37, 13
Decanoic acid	Lidocaine	4:1	37, 13
Menthol	Lidocaine	2:1	41
Thymol	Lidocaine	1:1	41
Thymol	Lidocaine	2:1	18, 37, 41
Decanoic acid	Atropine	2:1	41
Dodecanoic acid	Lidocaine	2:1	37, 41
Dodecanoic acid	Atropine	2:1	41
Thymol	Atropine	2:1	41
Decanoic acid	Trioctylphosphine oxide	1:2	42
Decanoic acid	Trioctylphosphine oxide	1:1	42
Decanoic acid	Dodecyl-methyl-sulfoxide	1:2	42
N,N′-dihexylthiourea	Trioctylphosphine oxide	1:1	42
N,N′-dihexylthiourea	Trioctylphosphine oxide	1:2	42
Dodecanoic acid	Trioctylphosphine oxide	1:1	42
1,2-Decanediol	Trioctylphosphine oxide	1:1	42
3,5-Ditertbutylcatechol	Trioctylphosphine oxide	2:1	42
3,5-Ditertbutylcatechol	Trioctylphosphine oxide	1:1	42
3,5-Ditertbutylcatechol	Dodecyl-methyl-sulfoxide	1:1	42
Bisphenol Z	Trioctylphosphine oxide	1:2	42
Decanoic acid	Aliquat 336	2:1	43
Oleic acid	Aliquat 336	2:1	43
Ibuprofen	Aliquat 336	3:7	43
Decanoic acid	Choline chloride	1:1	44
Decanoic acid	Choline chloride	2:1	44
Decanoic acid	Choline chloride	3:1	44
Phenylethyl alcohol	Choline chloride	4:1	45, 46
Methanoic acid	(−)-Menthol	1:1	25
Acetic acid	(−)-Menthol	1:1	25
L(+)-Lactic acid	(−)-Menthol	1:1	25
Octanoic acid	(−)-Menthol	1:1	25

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Table 1. continued

Component 1	Component 2	Molar ratio	ref
Decanoic acid	(−)-Menthol	1:1	25
Octanoic acid	(−)-Menthol	0.45:0.55	47
Decanoic acid	(−)-Menthol	0.35:0.65	47
Dodecanoic acid	(−)-Menthol	0.25:0.75	47
Pyruvic acid	DL-Menthol	2:1	3, 20
Lactic acid	DL-Menthol	2:1	3, 48
Levulinic acid	DL- Menthol	1:1	20, 21
Acrylic acid	DL-Menthol	2:1	48
Acetic acid	DL-Menthol	1:1	3, 20, 21, 49
Acetic acid	DL-Menthol	2:1	48
Propionic acid	DL-Menthol	2:1	48
Butyric acid	DL-Menthol	1:1	20
Butyric acid	DL-Menthol	2:1	48
Valeric acid	DL-Menthol	2:1	48
Hexanoic acid	DL-Menthol	1:1	20
Octanoic acid	DL-Menthol	1:2	22
Octanoic acid	DL-Menthol	1:1	20−22
Octanoic acid	DL-Menthol	2:1	22, 48
Decanoic acid	DL-Menthol	1:2	22, 14
Decanoic acid	DL-Menthol	1:1	20, 22
Decanoic acid	DL-Menthol	2:1	22, 48
Dodecanoic acid	DL-Menthol	1:2	3, 20−22, 50
Dodecanoic acid	DL-Menthol	2:1	48
Oleic acid	DL-Menthol	2:1	48
Decanoic acid	Menthol	1:1	18, 37
Decanoic acid	Menthol	2:1	41
Decanoic acid	Thymol	1:1	18, 37, 41
Levulinic acid	L-Menthol	1:1	51
Levulinic acid	L-Menthol	2:3	51
Levulinic acid	L-Menthol	3:2	51
Levulinic acid	Thymol	1:1	51
Levulinic acid	Thymol	2:3	51
Levulinic acid	Thymol	3:2	51
Levulinic acid	Carvacrol	1:1	51
Levulinic acid	Carvacrol	2:3	51
Levulinic acid	Carvacrol	3:2	51
CnH(2n+1)OOH; n = 8, 10, 12, 14, 16, 18	Menthol and thymol	–	52
Octanoic acid	L(−)menthol and thymol were tested for a broad range of mixtures within the phase diagram to find the eutectic composition of this components.	–	53
Decanoic acid	L(−)menthol and thymol were tested for a broad range of mixtures within the phase diagram to find the eutectic composition of this components.	–	53
Dodecanoic acid	L(−)menthol and thymol were tested for a broad range of mixtures within the phase diagram to find the eutectic composition of this components.	–	53
Myristic acid	L(−)menthol and thymol were tested for a broad range of mixtures within the phase diagram to find the eutectic composition of this components.	–	53
Palmitic acid	L(−)menthol and thymol were tested for a broad range of mixtures within the phase diagram to find the eutectic composition of this components.	–	53
Stearic acid	L(−)menthol and thymol were tested for a broad range of mixtures within the phase diagram to find the eutectic composition of this components.	–	53
Thymol	Menthol	1:1	41, 54
Thymol	Menthol	1:2	41
Thymol	Coumarin	1:1	41
Thymol	Coumarin	2:1	41
1-Tetradecanol	Menthol	1:2	41, 55
Thymol	1,2-Decanediol	2:1	41
Menthol	1-Napthol	2:1	41
Methyltrioctylammonium chloride	Three phase system	–	56
Capryl alcohol	Three phase system	–	56
Octylic acid	Three phase system	–	56
1:2:3 ratio	Three phase system	–	56

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Table 1. continued

Component 1	Component 2	Molar ratio	ref
Ketoprofen	Menthol	A variation of ratios forming in situ a DES	57
Diclofenac	Menthol	A variation of ratios forming in situ a DES	57
Decanoic acid	Dodecanoate sodium salt	4:1	58
Thymol	Camphor	7:3	59
Thymol	Camphor	3:2	59
Thymol	Camphor	1:1	59, 54
Thymol	10-Undecylenic acid	7:3	59
Thymol	10-Undecylenic acid	3:2	59
Thymol	10-Undecylenic acid	1:1	59
Thymol	10-Undecylenic acid	1:2	59
Thymol	10-Undecylenic acid	1:3	59
Thymol	10-Undecylenic acid	1:4	59
Thymol	Decanoic acid	3:2	59
Thymol	Decanoic acid	1:1	59
Thymol	Decanoic acid	1:2	59
Thymol	Decanoic acid	1:3	59
Menthol	Camphor	1:1	54
Menthol	Borneol	7:3	54
Menthol	Sobrerol	0.95:0.05	54
Thymol	Sobrerol	7:3	54
Thymol	Borneol	1:1	54
Camphor	Sobrerol	7:3	54
Borneol	Sobrerol	6:4	54

for DESs.[9] An aspect of high importance is the fact that DESs are mixtures of two or more components and not one pure salt consisting of ions. Thus, treating DESs as the new generation of ILs makes no sense due to the large differences. Moreover, it is stated that it is debatable whether a combination of a solid and a liquid component is actually a DES; it might be a solvolysis instead. The biggest debate is on the eutectic composition of a DES, the deepness of the eutectic point, and how to define this. In the end, Martins et al. decided on the following definition: "*A deep eutectic solvent a mixture of two or more pure compounds for which the eutectic point temperature is below that of an ideal liquid mixture, presenting significant negative deviations from ideality. Additionally, the temperature depression should be such that the mixture is liquid at operating temperature for a certain composition range. Otherwise, a simpler term "eutectic solvent" could be used to describe mixtures that do not fulfill these criteria.*[9]" This is a solid start to further discuss the definition of (hydrophobic) DESs. Recently, a new class of DESs, type V, has been introduced into the literature.[11] This type V class makes the field of (hydrophobic) DESs a little bit more clear. Type V DESs are composed solely of molecular substances. More specifically, it is reported that a hydrogen bond acceptor undergoes hydrogen bonding with a hydroxyl compound that is attached to an aromatic ring, for example, a phenolic compound.[11]

In this perspective, all the combinations of two solid components presented in the literature as hydrophobic DESs are given. However, the term "hydrophobic" should also not be taken lightly, and it might be best to put some constraints on it too. Nowadays, components such as acetic acid and lactic acid have been presented as compounds of hydrophobic DESs, as can be read in this perspective, while they have relatively high solubility in water. In the literature, also the name NADES (natural deep eutectic solvents) is used on many occasions.[7] We think it should be a goal of the DES field itself to produce solvents with natural and safe components.

It is often stated that DESs have many advantages over conventional solvents, such as easy preparation, no need of purification, and tunability of properties.[4] These can, in general, be considered as true. For hydrophobic DESs, it can also be given that they are water immiscible, although for this property it should be mentioned that this mainly depends on the compounds and their hydrophobicity. Often DESs are also mentioned as biodegradable,[4] but one should not forget that this will be determined by the components that are used and their synergetic effect. The fact that (hydrophobic) DESs have a lower volatility is recently proven, but also for this property, it truly depends on the components that are used for a certain DES.[12]

The field of hydrophobic DESs is now almost five years old and has evolved during this time. It started with two publications in 2015 where the first hydrophobic DESs were presented and used for liquid−liquid extractions.[2,3] These were hydrophobic DESs that were far from ideal at that moment with high viscosities and large amounts of DES components that leached to the water phase. In 2016, still only a few papers were presented, but the very impressive work on the removal of metals from an aqueous environment was started.[13,14] The year 2017 was really a year of growth for the hydrophobic DES field. Some very interesting applications that were investigated include the extraction of Artemisinin, a medicine for malaria, from *Artemisia annua* leaves,[15] the first microextractions,[16] photon upconversion,[17] and the use of hydrophobic DESs in membranes.[18] In the years after, a rapid growth of the field was observed. Some interesting papers are, for example, related to theoretical approaches,[10] determination of the liquid phase behavior,[19] and the introduction of the type V DESs, specifically introduced for hydrophobic DESs.[11]

A full overview of all hydrophobic DESs presented so far in the literature is shown in Table 1. This impressive list built in approximately 4−5 years in combination with the applications already tested shows the potential of hydrophobic DESs for

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Table 2. Density (ρ) at Temperature of 298.15 K of Hydrophobic DESs

Component 1	Component 2	Ratio	ρ (g·cm^{-3})	ref
Menthol	Decanoic acid	1:1	0.910	60
Menthol	Octanoic acid	1:1	0.910	60
Menthol	Oleic acid	1:1	0.910	60
Menthol	Salol	1:1	1.070	60
Menthol	Acetic acid	1:1	0.931	3
Menthol	Pyruvic acid	1:2	0.995	3
Menthol	Lactic acid	1:2	1.033	3, 62
Menthol	dodecanoic acid	2:1	0.894	3
Menthol	Decanoic acid	1:1	0.896	62
Menthol	Levulinic acid	1:1	0.985	63
Menthol	Lidocaine	2:1	0.939	41
Dodecanoic acid	Octanoic acid	1:3	0.901	64
Dodecanoic acid	Nonanoic acid	1:3	0.897	64
Dodecanoic acid	Decanoic acid	1:2	0.894	64
Dodecanoic acid	Lidocaine	2:1	0.950	41
Dodecanoic acid	Atropine	2:1	1.009	41
Lidocaine	Decanoic acid	1:2	0.958	13
Lidocaine	Decanoic acid	1:3	0.950	13
Lidocaine	Decanoic acid	1:4	0.942	13
tetrabutylammonium chloride	Decanoic acid	1:2	0.917	2
Tetrabutylammonium bromide	Octanoic acid	1:2	0.974	65
Tetrabutylammonium bromide	Decanoic acid	1:2	0.957	65
Tetrabutylammonium bromide	Oleic acid	1:2	0.959	65
Methyltrioctylammonium chloride	Decanoic acid	1:2	0.896	2
Methyltrioctylammonium chloride	Ethyl 4-hydroxybenzoate	1:2	0.995	66
Methyltrioctylammonium chloride	Hydroquinone	1:1	0.952	32
Methyltrioctylammonium chloride	4-phenylphenol	1:1	0.937	32
Methyltrioctylammonium chloride	4-cyanophenol	2:1	0.919	32
Methyltrioctylammonium chloride	4-cyanophenol	1:1	0.941	32
Methyltrioctylammonium chloride	4-cyanophenol	1:2	0.974	32
Tetraoctylammonium chloride	Decanoic acid	1:2	0.889	2
	Decanoic acid	1:1.5	0.888	68
Methyltrioctylammonium bromide	Decanoic acid	1:2	0.942	2
Tetraoctylammonium bromide	Decanoic acid	1:2	0.930	2
Tetraheptylammonium chloride	Decanoic acid	1:2	0.891	2
Thymol	Camphor	7:3	0.970	59
Thymol	Camphor	3:2	0.973	59
Thymol	Camphor	1:1	0.987	59
Thymol	10-Undecylenic acid	7:3	0.960	59
Thymol	10-Undecylenic acid	3:2	0.952	59
Thymol	10-Undecylenic acid	1:1	0.946	59
Thymol	10-Undecylenic acid	1:2	0.940	59
Thymol	10-Undecylenic acid	1:3	0.938	59
Thymol	10-Undecylenic acid	1:4	0.935	59
Thymol	Decanoic acid	3:2	0.952	59
Thymol	Decanoic acid	1:1	0.944	59
Thymol	Decanoic acid	1:2	0.937	59
Thymol	Decanoic acid	1:3	0.927	59
Thymol	Lidocaine	2:1	0.989	41
Thymol	Lidocaine	1:1	0.993	41
Thymol	Coumarin	2:1	1.050	41
Thymol	Coumarin	1:1	1.092	41
Thymol	Menthol	1:1	0.937	41
Thymol	Menthol	1:2	0.924	41
Trioctylphosphine Oxide	Phenol	1:2	0.933	69
Betaine	Hexafluoroisopropanol	1:2	1.476	70
Betaine	Hexafluoroisopropanol	1:2.5	1.490	70
Betaine	Hexafluoroisopropanol	1:3	1.525	70
L-Carnitine	Hexafluoroisopropanol	1:2	1.503	70
L-Carnitine	Hexafluoroisopropanol	1:2.5	1.542	70
L-Carnitine	Hexafluoroisopropanol	1:3	1.505	70

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Table 2. continued

Component 1	Component 2	Ratio	ρ (g·cm^{-3})	ref
N,N-dimethyl-N,N-didodecylammonium chloride	Thymol	1:2	0.880	71
N,N-dimethyl-N,N-didodecylammonium chloride	Phenylacetic acid	1:2	0.940	71
N,N-dimethyl-N,N-didodecylammonium chloride	Glycolic acid	1:3	1.040	71
Trimethylglycine	Thymol	1:3	0.960	71
Trimethylglycine	Phenylacetic acid	1:2	1.160	71
Trimethylglycine	Glycolic acid	1:2	1.270	71
Decanoic acid	lidocaine	2:1	0.961	41
Decanoic acid	Atropine	2:1	1.027	41
Decanoic acid	Menthol	1:1	0.900	41
Decanoic acid	Menthol	1:2	0.900	41
1-tetradecanol	Menthol	1:2	0.872	41
1,2-decanediol	Thymol	1:2	0.952	41
1-Napthol	Menthol	1:2	0.971	41
Atropine	Thymol	1:2	1.062	41
Menthol	Thymol	1:1	0.9327	54
Menthol	Camphor	1:1	0.9237	54
Menthol	Borneol	7:3	0.9149	54
Thymol	Camphor	1:1	0.9668	54

future use in academics and industry. As presented in the previous paragraph, they have the potential to become truly innovative and sustainable solvents. However, it should be mentioned that this will all depend on the components chosen, and its potential will most likely also depend on the application that can be performed with them.

PHYSICOCHEMICAL PROPERTIES OF HYDROPHOBIC DESS

The physicochemical properties and thermal stabilities of all reported hydrophobic DESs so far are presented in the following sections.

Density. Density (ρ) is one of the essential properties for solvent design and applications. In principle, the densities of DESs are related to the molecular packing and the interactions between its constituents. Table 2 lists the density data of the hydrophobic DESs reported in the literature. From Table 2, it can be observed that the hydrophobic DESs show a density between 0.88 and 1.5 g·cm^{-3}. A proportional relationship of the density and its constituents is evident for the components of the DESs at the molar ratios. As an example, for menthol-based DESs at molar ratio (1:1), the hydrophobic DESs densities followed the order menthol:salol > menthol:acetic acid > menthol:octanoic acid > menthol:decanoic acid, which is the same order of the component 2's densities. However, no final conclusion can be drawn, and further experimental measurements are needed for a broader range of DES combinations at the same molar ratios. Furthermore, for the same component 2 and molar ratio, the density of the hydrophobic DESs decreases with increasing the alkyl chain length of component 1. The density of ammonium-based DESs in a molar ratio of (component 1:component 2 = 1:2) follows the order tetrabutylammonium chloride:decanoic acid > tetraheptylammonium chloride:decanoic acid > tetraoctylammonium chloride:decanoic acid. Furthermore, van Osch et al. have tested the change in densities after mixing with water. The results showed a slight influence of the DES density due to the low water uptake.[41] Additionally, a linear decrease of the DES density with temperature was reported by Ribeiro et al.[3]

Viscosity. The viscosity of the solvents is one of the determinant factors in its selection especially for an industrial-scale application. It is expected that DESs are formed via hydrogen bonding interactions, limiting the mobility of the DESs molecules; therefore, high DES viscosities are expected. However, viscosities as low as 7 mPa·s were reported for the hydrophobic DES dodecanoic acid:octanoic acid (1:3) (Table 2). All the viscosities of the hydrophobic DESs reported so far are listed in Table 3. It can be seen that the hydrophobic DESs have a large range of viscosities (~7−86,800 mPa·s). This allows big room for task-oriented solvent design. The chemical nature and the interactions between the DES constituents are the main factors affecting the viscosity. It should be mentioned that no straightforward relation can be found in viscosities for the same HBA with different HBDs at fixed molar ratio. Moreover, it was observed that the viscosity changes significantly with temperature. The increase in temperature leads to a substantial decrease in viscosity following an Arrhenius-like behavior.

Melting Point. The melting point of a DES is the lowest temperature at which the solvent could be applied in liquid state. By general definition, the DESs exhibit a melting point that is far below that of its individual constituents. Table 4 lists the melting temperatures (T_m) of hydrophobic DESs reported in the literature. It can be observed that the melting points of all hydrophobic DESs studied are well below room temperature. Few hydrophobic DESs (i.e., menthol:pyruvic acid and menthol:dodecanoic acid) exhibited two melting points; however, both temperatures are well below that of both constituents (Table 4). The melting point is highly dependent on the structure and strength of the interactions between the HBD and HBA.

A problem within the field of hydrophobic DES is the presumption of melting points. Often a ratio between two solid points is considered as the eutectic point as soon as the combination of two solid components becomes liquid. However, the melting points should be determined by investigating the full solid−liquid phase behavior to find its true eutectic point. Recently, new investigations showed the use of phase diagrams to determine the composition at the eutectic point and its melting point. For example, phase diagrams of menthol with octanoic acid, decanoic acid, dodecanoic acid, myristic acid, palmitic acid, and stearic acid were tested.[53] This is the way forward for the field of hydrophobic DESs. Measuring

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Table 3. Viscosity (η) at Temperature of 298.15 K of Hydrophobic DESs

Component 1	Component 2	Ratio	η(mPa·s)	ref
Menthol	Acetic acid	1:1	8.691	63
Menthol	Pyruvic acid	1:2	29.951	63
Menthol	Lactic acid	1:2	218.927	63
Menthol	Dodecanoic acid	2:1	24.417	63
Menthol	Levulinic acid	1:1	30.1	63
Menthol	Lidocaine	2:1	68.05	41
Dodecanoic acid	Octanoic acid	1:3	7.085	64
Dodecanoic acid	Nonanoic acid	1:3	8.636	64
Dodecanoic acid	Decanoic acid	1:2	10.756	64
Dodecanoic acid	Lidocaine	2:1	370.55	41
Dodecanoic acid	Atropine	2:1	5599.5	41
Lidocaine	Decanoic acid	1:2	237.5	13
Lidocaine	Decanoic acid	1:3	208.5	13
Lidocaine	Decanoic acid	1:4	142	13
Tetrabutylammonium chloride	Decanoic acid	1:2	265.26	2
Methyltrioctylammonium chloride	Decanoic acid	1:2	783.41	2
Methyltrioctylammonium chloride	Ethyl 4-hydroxybenzoate	1:2	957.5	66
Tetraoctylammonium chloride	Decanoic acid	1:2	472.58	2
Methyltrioctylammonium bromide	Decanoic acid	1:2	576.53	2
Tetraoctylammonium bromide	Decanoic acid	1:2	636.36	2
Tetraheptylammonium chloride	Decanoic acid	1:2	172.87	2
Thymol	Camphor	7:3	18.8	59
Thymol	Camphor	3:2	20.5	59
Thymol	Camphor	1:1	25.8	59
Thymol	10-Undecylenic acid	7:3	15.6	59
Thymol	10-Undecylenic acid	3:2	14.4	59
Thymol	10-Undecylenic acid	1:1	13.2	59
Thymol	10-Undecylenic acid	1:2	13.1	59
Thymol	10-Undecylenic acid	1:3	12.4	59
Thymol	10-Undecylenic acid	1:4	11.8	59
Thymol	Decanoic acid	3:2	13.0	59
Thymol	Decanoic acid	1:1	11.2	59
Thymol	Decanoic acid	1:2	10.8	59
Thymol	Decanoic acid	1:3	10.4	59
Thymol	Lidocaine	2:1	122.05	41
Thymol	Lidocaine	1:1	177.15	41
Thymol	Coumarin	2:1	31.35	41
Thymol	Coumarin	1:1	29.16	41
Thymol	Menthol	1:1	53.14	41
Thymol	Menthol	1:2	67.85	41
Trioctylphosphine Oxide	Phenol	1:2	12.38	69
Betaine	Hexafluoroisopropanol	1:2	76	70
Betaine	Hexafluoroisopropanol	1:2.5	55	70
Betaine	Hexafluoroisopropanol	1:3	46	70
L-Carnitine	Hexafluoroisopropanol	1:2	698	70
L-Carnitine	Hexafluoroisopropanol	1:2.5	194	70
L-Carnitine	Hexafluoroisopropanol	1:3	149	70
Decanoic acid	Lidocaine	2:1	360.6	41
Decanoic acid	Atropine	2:1	5985.0	41
Decanoic acid	Menthol	1:1	20.03	41
Decanoic acid	Menthol	1:2	27.67	41
1-Tetradecanol	Menthol	1:2	43.86	41
1,2-Decanediol	Thymol	1:2	64.25	41
1-Napthol	Menthol	1:2	120.9	41
Atropine	Thymol	1:2	86800	41
Menthol	Thymol	1:1	38.08	54
Menthol	Camphor	1:1	16.42	54
Menthol	Borneol	7:3	110.40	54
Thymol	Camphor	1:1	20.80	54

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Table 4. Melting Point of Hydrophobic DESs

Component 1	Component 2	Ratio	T_m[K]	ref
Menthol	Octanoic acid	1:1	269.73	60
Menthol	Oleic acid	1:1	272.64	60
Menthol	Salol	1:1	295.43	60
Menthol	Acetic acid	1:1	265.34	3
Menthol	Pyruvic acid	1:2	214.34 and 266.37	3
Menthol	Lactic acid	1:2	212.01	3
Menthol	Dodecanoic acid	2:1	280.28, 286.99	3
Menthol	Decanoic acid	1:1	285.69	60
Dodecanoic acid	Octanoic acid	1:3	282.15	64
Dodecanoic acid	Nonanoic acid	1:3	282.15	64
Dodecanoic acid	Decanoic acid	1:2	291.15	64
Tetrabutylammonium chloride	Decanoic acid	1:2	261.2	2
Tetrabutylammonium bromide	Octanoic acid	1:2	273.73	65
Tetrabutylammonium bromide	Decanoic acid	1:2	289.15−290.15	65
Tetrabutylammonium bromide	Oleic acid	1:2	275.01	65
Methyltrioctylammonium chloride	Decanoic acid	1:2	273.1	2
Methyltrioctylammonium chloride	Ethyl 4-hydroxybenzoate	1:2	210.15	66
Methyltrioctylammonium chloride	Hydroquinone	1:1	195.15	67
Methyltrioctylammonium chloride	4-phenylphenol	1:1	301.15	67
Methyltrioctylammonium chloride	4-cyanophenol	2:1	210.15	67
Methyltrioctylammonium chloride	4-cyanophenol	1:1	211.15	67
Methyltrioctylammonium chloride	4-cyanophenol	1:2	210.15	67
Tetraoctylammonium chloride	Decanoic acid	1:2	275.1	2
Methyltrioctylammonium bromide	Decanoic acid	1:2	282.1	2
Tetraoctylammonium bromide	Decanoic acid	1:2	282.1	2
Tetraheptylammonium chloride	Decanoic acid	1:2	256.5	2
Tetraheptylammonium chloride	Ibuprofen	7:3	200.33	14
Tetraheptylammonium chloride	Oleic acid	1:2	231.95, 238.46, 262.8	14
Thymol	Camphor	7:3	240.15	59
Thymol	Camphor	3:2	236.15	59
Thymol	Camphor	1:1	229.15	59
Thymol	10-Undecylenic acid	7:3	292.15	59
Thymol	10-Undecylenic acid	3:2	289.65	59
Thymol	10-Undecylenic acid	1:1	284.15	59
Thymol	10-Undecylenic acid	1:2	283.15	59
Thymol	10-Undecylenic acid	1:3	282.15	59
Thymol	10-Undecylenic acid	1:4	280.65	59
Thymol	Decanoic acid	3:2	291.15	59
Thymol	Decanoic acid	1:1	290.15	59
Thymol	Decanoic acid	1:2	291.15	59
Thymol	Decanoic acid	1:3	292.15	59
Trioctylphosphine Oxide	Phenol	1:2	240.15	69
Betaine	Hexafluoroisopropanol	1:2	233.75	70
Betaine	Hexafluoroisopropanol	1:2.5	239.85	70
Betaine	Hexafluoroisopropanol	1:3	238.45	70
L-Carnitine	Hexafluoroisopropanol	1:2	254.45	70
L-Carnitine	Hexafluoroisopropanol	1:2.5	236.65	70
L-Carnitine	Hexafluoroisopropanol	1:3	255.95	70
N,N-dimethyl-N,N-didodecylammonium chloride	Thymol	1:2	253.15	71
N,N-dimethyl-N,N-didodecylammonium chloride	Phenylacetic acid	1:2	266.15	71
N,N-dimethyl-N,N-didodecylammonium chloride	Glycolic acid	1:3	278.15	71
Trimethylglycine	Thymol	1:3	258.15	71
Trimethylglycine	Phenylacetic acid	1:2	266.15	71
Trimethylglycine	Glycolic acid	1:2	237.15	71

the melting point of a certain molar ratio without knowledge of the phase diagram is simply not going to cut it. The solid−liquid diagram should always be determined of any combination of solids.

Degradation Temperature. The knowledge of the thermal stability range is important to assess the viability of a solvent. Within this range, the DES stays intact and in a liquid state. Since DESs decompose before reaching the critical temperature, the highest operating limit of the DESs is characterized by the

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degradation temperature (T_{deg}), while the lowest operating limit is characterized by the melting point (T_m). Table 5 lists the degradation temperatures of the hydrophobic DESs. Like the melting point of DESs, the degradation behavior of DESs is highly dependent on the nature of the DES constituents.

Table 5. Degradation Temperatures of Hydrophobic DESs

Component 1	Component 2	Ratio	T_{deg}[K]	ref
Menthol	Acetic acid	1:1	473.94	3
Menthol	Pyruvic acid	1:2	491.65	3
Menthol	Lactic acid	1:2	502.04	3
Menthol	Dodecanoic acid	2:1	504.64	3
Tetrabutylammonium chloride	Decanoic acid	1:2	467.79	2
Methyltrioctylammonium chloride	Decanoic acid	1:2	472.41	2
Tetraoctylammonium chloride	Decanoic acid	1:2	480.81	2
Methyltrioctylammonium bromide	Decanoic acid	1:2	459.15	2
Tetraoctylammonium bromide	Decanoic acid	1:2	459.31	2
Tetraheptylammonium chloride	Decanoic acid	1:2	472.37	2
Thymol	Lidocaine	2:1	412.9	41
Thymol	Lidocaine	1:1	392.9	41
Thymol	Coumarin	2:1	390.8	41
Thymol	Coumarin	1:1	381.9	41
Thymol	Menthol	1:1	392.9	41
Thymol	Menthol	1:2	378.7	41
Decanoic acid	Lidocaine	2:1	443.1	41
Decanoic acid	Atropine	2:1	444.8	41
Decanoic acid	Menthol	1:1	410.2	41
Decanoic acid	Menthol	1:2	382.9	41
1-tetradecanol	Menthol	1:2	386.3	41
1,2-decanediol	Thymol	1:2	395.7	41
1-Napthol	Menthol	1:2	388.5	41
Atropine	Thymol	1:2	429.6	41
Dodecanoic acid	Lidocaine	2:1	459.8	41
Dodecanoic acid	Atropine	2:1	477	41

Volatilities. Recently, a new physicochemical property of hydrophobic DESs was determined, namely, volatility.[12] The total vapor pressure of six hydrophobic DESs was tested, namely, decanoic acid:thymol (1:1), decanoic acid:lidocaine (2:1), decanoic acid:lidocaine (3:1), decanoic acid:lidocaine (4:1), decanoic acid:menthol (1:1), and thymol:lidocaine in a 2:1 molar ratio. The most volatile DES was the one composed of decanoic acid and menthol, with a volatility that is 150−1000 times smaller than toluene in the temperature region of 320−380 K.[12] This research with the first preliminary results on the volatilities shows that volatility is an important factor to further investigate in the future for hydrophobic and hydrophilic DESs.

Solvatochromic Properties. In 2018, intermolecular interactions by using solvatochromic probes for hydrophobic and hydrophilic DESs was explored.[21] The hydrophobic DESs investigated were levulinic acid:tetrabutylammonium chloride (2:1), octanoic acid:tetrabutylammonium chloride (2:1), decanoic acid:tetrabutylammonium chloride (2:1), dodecanoic acid:tetrabutylammonium chloride (2:1), levulinic acid:DL-Menthol (1:1), octanoic acid:DL-Menthol (1:1), dodecanoic acid:DL-Menthol (1:2), and acetic:DL-Menthol (1:1). The intermolecular interactions were measured with the Kamlet−Taft parameters. The Kamlet−Taft α, β, π^*, and E_TN were evaluated. Overall, it was found that the DESs have a high hydrogen bond acidity, caused by the acids, and not much variation in hydrogen bonding basicity if the same compound as HBA was used. As expected, substantial differences were found in hydrogen bonding basicity when the HBA was varied. Of the tested DESs, the HBA controlled the dipolarity/polarizability. Surprisingly, DESs showed higher hydrogen bonding donating and accepting ability in comparison to ILs and conventional solvents. Furthermore, DESs composed of a combination between the acids octanoic, decanoic, dodecanoic, myristic, palmitic, and stearic acids and the terpenes L(−)-menthol and thymol were tested.[53] Also here, solvatochromic parameters were investigated with the Kamlet−Taft parameters, from which it could be concluded that the chosen terpene had the biggest influence on the parameters. Moreover, thymol induced a higher hydrogen bond acidity, while menthol increased the hydrogen bond basicity. Recently, the Kamlet−Taft parameters of the combinations of menthol + thymol, menthol + camphor, menthol + borneol, menthol + sobrerol, thymol + camphor, thymol + borneol, and thymol + sobrerol were studied.[54] The results showed that thymol has a higher polarizability/dipolarity due to the presence of its aromatic ring. Moreover, it was found that there is a variation between the hydrogen bond donating and accepting ability. Really interesting was the tunability of the α, β, and π^* behavior found for menthol + thymol and thymol + camphor combinations. An overview of the Kamlet−Taft parameters presented in the literature is given in Table 6.

APPLICATIONS OF HYDROPHOBIC DESS

Table 7 gives an overview of all the applications of hydrophobic DESs as found in the literature with their references. All the applications are further discussed in the paragraphs given below. Many of the presented applications are liquid−liquid extractions, which form quite complicated systems. The hydrophobic DES itself is already a binary or a ternary system. With the addition of an aqueous phase and a solute, quaternary or quinary systems are formed, which tend to have all kinds of interactions and thermodynamic effects.

Removal of Volatile Fatty Acids from Water. Hydrophobic DESs were discovered in 2015.[2,3] Despite the field of hydrophobic DESs being relatively young, some applications have been presented in the literature. The first application tested for hydrophobic DESs was the removal of volatile fatty acids (VFAs) from water. Hydrophobic DESs based on decanoic acid and a variation of quaternary ammonium salts were used as solvent, while the VFAs acetic, propionic, and butyric acids were used as pollutants of the water phase.[2] The results are depicted in Table 8. They show that all developed hydrophobic DESs outperform the industrial standard trioctylamine and that with an increase in the chain length of the VFA the extraction efficiency increases. More specifically, decanoic acid:methyltrioctylammonium bromide (2:1) had the highest extraction efficiency for the VFAs, most likely since the methyl group induces lower steric hindrance. An exception to these results was found for propionic acid and butyric acid for the DES decanoic acid:tetraheptylammonium chloride (2:1). Most likely, the extraction efficiencies of this DES were higher due to turbidities in the water phase after centrifugation, an indication that no proper phase separation occurred. Moreover, the results showed that a change from bromide to chloride anion gave a decrease in the extraction efficiency, caused by an increase in steric hindrance.

In 2019, another investigation was presented in the literature where hydrophobic DESs were used as VFA extractants.[42] An impressive amount of 100 combinations of 16 hydrophobic components was tested, of which mixtures of dihexylthiourea

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Table 6. Kamlet−Taft Parametes of Hydrophobic DESs

Component 1	Component 2	x1	α	β	π^*	ref
Levulinic acid	tetrabutylammonium chloride (2:1)	0.67	*	0.82	1.06	21
Octanoic acid	tetrabutylammonium chloride (2:1)	0.67	1.41	0.99	0.76	21
Decanoic acid	tetrabutylammonium chloride (2:1)	0.67	1.36	0.97	0.73	21
Dodecanoic acid	tetrabutylammonium chloride (2:1)	0.67	1.45	1.04	0.71	21
Levulinic acid	DL-Menthol	0.5	1.56	0.58	0.66	21
Octanoic acid	DL-Menthol	0.5	1.77	0.50	0.41	21
Dodecanoic acid	DL-Menthol	0.33	1.79	0.57	0.37	21
Acetic acid	DL-Menthol	0.5	1.64	0.60	0.53	21
L(−)-Menthol	Octanoic acid	0.43	0.85	0.43 ± 0.01	0.39 ± 0.01	53
L(−)-Menthol	Decanoic acid	0.57	0.84	0.45 ± 0.01	0.35 ± 0.01	53
L(−)-Menthol	Dodecanoic acid	0.68	0.79	0.54 ± 0.02	0.37 ± 0.01	53
L(−)-Menthol	1-tetradecanoic acid	0.78	0.75	0.50 ± 0.02	0.38 ± 0.01	53
L(−)-Menthol	Hexadecanoic acid	0.85	0.71	0.57 ± 0.01	0.38 ± 0.01	53
L(−)-Menthol	Octadecanoic acid	0.91	0.68	0.64 ± 0.02	0.38 ± 0.01	53
Thymol	Octanoic acid	0.29	1.10	0.05 ± 0.02	0.67 ± 0.01	53
Thymol	Decanoic acid acid	0.43	1.11	0.05 ± 0.02	0.71 ± 0.01	53
Thymol	Dodecanoic acid	0.55	1.05	0.02 ± 0.01	0.75 ± 0.01	53
Thymol	1-tetradecanoic acid	0.67	1.13	0.02 ± 0.02	0.84 ± 0.01	53
Thymol	Hexadecanoic acid	0.76	1.11	0.01 ± 0.01	0.87 ± 0.01	53
Thymol	Octadecanoic acid	0.84	1.10	0.05 ± 0.01	0.94 ± 0.01	53
Menthol	Thymol	0.10	1.00	0.15 ± 0.01	0.97 ± 0.01	54
Menthol	Thymol	0.20	0.97	0.18 ± 0.01	0.90 ± 0.01	54
Menthol	Thymol	0.30	0.92	0.22 ± 0.01	0.84 ± 0.01	54
Menthol	Thymol	0.40	0.88	0.23 ± 0.01	0.83 ± 0.01	54
Menthol	Thymol	0.50	0.84	0.28 ± 0.02	0.77 ± 0.01	54
Menthol	Thymol	0.60	0.79	0.32 ± 0.01	0.72 ± 0.01	54
Menthol	Thymol	0.70	0.70	0.36 ± 0.01	0.67 ± 0.01	54
Menthol	Thymol	0.80	0.64	0.48 ± 0.01	0.59 ± 0.01	54
Menthol	Thymol	0.90	0.56	0.53 ± 0.01	0.53 ± 0.01	54
Menthol	Camphor	0.50	0.41	0.61 ± 0.01	0.52 ± 0.01	54
Menthol	Borneol	0.70	0.53	0.63 ± 0.01	0.43 ± 0.01	54
Menthol	Sorbrerol	0.95	0.52	0.68 ± 0.02	0.43 ± 0.01	54
Thymol	Camphor	0.30	0.68	0.45 ± 0.01	0.59 ± 0.01	54
Thymol	Camphor	0.40	0.71	0.41 ± 0.01	0.62 ± 0.01	54
Thymol	Camphor	0.50	0.82	0.34 ± 0.02	0.69 ± 0.02	54
Thymol	Camphor	0.60	0.88	0.26 ± 0.01	0.77 ± 0.01	54
Thymol	Camphor	0.70	0.94	0.23 ± 0.01	0.85 ± 0.01	54
Thymol	Camphor	0.80	0.97	0.18 ± 0.06	0.93 ± 0.01	54
Thymol	Camphor	0.90	1.01	0.17 ± 0.01	0.98 ± 0.01	54
Thymol	Borneol	0.50	0.87	0.27 ± 0.01	0.78 ± 0.01	54
Thymol	Sobrerol	0.70	0.99	0.11 ± 0.04	0.94 ± 0.01	54

were stable in a wide range of pH and gave the best extraction efficiencies. It was shown that only undissociated acids were removed from the water phase, and an increase of extraction efficiency was achieved when the hydrophobicity of the VFAs was increased.

Extraction of Biomolecules, Pesticides, and Medicinal Components from Aqueous Solutions. In 2015, the group of Marrucho presented hydrophobic DESs based on DL-menthol and acids, i.e., DL-menthol:acetic acid (1:1), DL-menthol:pyruvic acid (1:2), DL-menthol:lactic acid (1:2), and DL-menthol:dodecanoic acid (2:1).[3] These hydrophobic DESs were used for the removal of different biomolecules from aqueous solutions, except for the one with pyruvic acid since it has the presence of an absorption band at about 350 nm. The pHs at which the extraction occurred is added to Table 9. The results show that reasonable high partition coefficients were achieved for the extraction of caffeine and tryptophan from water with the hydrophobic DES DL-menthol:lactic acid at a pH of 1.18. For vanillin, moderate partition coefficients were found for all the hydrophobic DESs, while for isopthalic acid the highest partition coefficient was achieved with the DES composed of DL-menthol:lactic acid at a pH of 1.47.

Hydrophobic DESs based on fatty acids were used as solvent for the extraction of bisphenol A from the water phase.[64] The three DESs in their investigation were octanoic acid:dodecanoic acid (3:1), nonanoic acid:dodecanoic acid (3:1), and decanoic acid:dodecanoic acid (2:1), for which melting points of 9, 9, and 18 °C were found, respectively. The extraction efficiencies of bisphenol A from the water phase were 76.04% for the octanoic acid:dodecanoic acid (3:1), 88.32% for the nonanoic acid:-dodecanoic acid (3:1), and) 81.81% for the decanoic acid:-dodecanoic acid (2:1). After testing these binary mixtures, also ternary mixtures were prepared and tested for the extraction of bisphenol A from the aqueous phase. The ternary DESs were octanoic acid:nonanoic acid:dodecanoic acid (1:1:1), octanoic acid:nonanoic acid:dodecanoic acid (1:2:1), octanoic acid:-

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Table 7. Application Tested with Hydrophobic DESs

Application	ref
Removal of fatty acid from water	2, 42
Extraction of biomolecules, pesticides, and medicinal components from aqueous solutions	3, 64, 37, 20, 28, 41
Metal ion removal from aqueous solutions	14, 13, 52, 40, 35, 69, 27
Other liquid−liquid extractions	50, 59, 71, 62
Liquid−liquid microextraction	23, 57, 45, 46
Vortex-assisted liquid microextraction	67, 33
Dispersive liquid−liquid microextraction	16, 22, 31, 43, 70, 39
Microextractions based on the solidification of floating drops	65, 60, 44
Ultrasound-assisted emulsification liquid phase microextraction	72, 73
Shaker-assisted liquid−liquid microextraction	34
Formation of a two phase system of a hydrophilic and a hydrophobic DES	56, 25
Three phase systems	51
Removal of components from leaves	15, 30, 24, 61
Solubility of CO_2 in hydrophobic DESs	68, 36,78
Formation of hydrogels	58
Formation of oil in water emulsions with hydrophobic DESs	55
Hydrophobic DESs and their use in wound healing	74
Membrane formation	26, 18
Centrifugal partition chromatography	63
Formation of a ferrofluid	48
Coatings	66
Photo luminescence and dye sensitized solar cells	17, 29, 49
Catalysis	38, 47

Table 8. Extraction Efficiencies of Acetic, Propionic, and Butyric Acids from Diluted Aqueous Solutions with Reported DESs and Trioctylamine (TOA)

	E [%]		
DESs	Acetic acid	Propionic acid	Butyric acid
Decanoic acid:methyltrioctylammonium chloride (2:1)	38.0	70.5	89.8
Decanoic acid:tetraheptylammonium chloride (2:1)	32.0	76.5[a]	91.5[a]
Decanoic acid:tetraoctylammonium chloride (2:1)	25.0[a]	52.7	81.3
Decanoic acid:methyltrioctylammonium bromide (2:1)	29.7	63.4	83.1
Decanoic acid:tetraoctylammonium bromide (2:1)	30.6	65.9	87.4
Trioctylamine	18.6	45.9	73.5

[a]The top phase (DES phase) was turbid.

nonanoic acid:dodecanoic acid (2:1:1), octanoic acid:nonanoic acid:dodecanoic acid (3:1:1), octanoic acid:nonanoic acid:-dodecanoic acid (3:2:1), nonanoic acid:decanoic acid:dodecanoic acid (1:1:1), nonanoic acid:decanoic acid:dodecanoic acid (1:2:1), nonanoic acid:decanoic acid:dodecanoic acid (2:1:1), nonanoic acid:decanoic acid:dodecanoic acid (2:2:1), nonanoic acid:decanoic acid:dodecanoic acid (3:1:1), nonanoic acid:-decanoic acid:dodecanoic acid (3:2:1), octanoic acid:decanoic acid:dodecanoic acid (1:1:1), octanoic acid:decanoic acid:-dodecanoic acid (2:1:1), octanoic acid:decanoic acid:dodecanoic acid (3:1:1), and octanoic acid:decanoic acid:dodecanoic acid (3:2:1). The DES nonanoic acid:decanoic acid:dodecanoic acid (3:1:1) performed the best with an extraction efficiency of 91.52%.

Eight hydrophobic DESs were tested for their water and 5-hydroxymethylfurfural (HMF) solubility.[37] The tested hydrophobic DESs were decanoic acid:n-tetraoctylammonium bromide (2:1), decanoic acid:thymol (1:1), decanoic acid:lidocaine (2:1), decanoic acid:lidocaine (3:1), decanoic acid:lidocaine (4:1), decanoic acid:menthol (1:1), thymol:lidocaine (2:1), and dodecanoic acid:lidocaine (2:1). Water contents in these DESs after equilibrium with water varied between 1.72 and 17.70 wt %. The hydrophobic DESs decanoic acid:thymol (1:1), decanoic acid:n-tetraoctylammonium bromide (2:1), and thymol:lidocaine (2:1) had a HMF solubility of 80, 75, and 85 wt %, respectively.

The group of Marrucho worked on the use of hydrophobic DESs for the removal of pesticides from an aquatic environment.[20] More specifically, the four neonicotinoids Imidacloprid, Acetamiprid, Nitenpyram and Thiamethoxam were investigated. The DESs were a combination of the HBAs DL-menthol and tetrabutylammonium chloride and the organic acids, acetic acid, pyruvic acid, butyric acid, levulinic acid, butyric acid, hexanoic acid, octanoic acid, decanoic acid, and dodecanoic acid. They chose the stability in the water phase as an important factor, which led to the conclusion that the combination of octanoic acid with DL-menthol in a 1:1 molar ratio, decanoic acid with DL-menthol in a 1:1 molar ratio, and dodecanoic acid with DL-menthol in a 1:2 molar ratio were the most useful combinations. The results showed that extraction efficiencies up to 80% could be achieved. A clear inverse order of the water solubility of the proteins and their extraction efficiency was observed. No clear trend was found between the DES hydrophobicity and the extraction efficiency.

Hydrophobic DESs based on fatty acid/alcohol were used to replace conventional organic solvents for the removal of antibiotics from environmental water with a liquid−liquid microextraction as method.[28] The antibiotics used in the studies are levofloxacin and ciprofloxacin. A list of 14 hydrophobic DESs in a 1:1 molar ratio were investigated, i.e., hexanoic acid:-tetrabutylammonium bromide, octanoic acid:tetrabutylammonium bromide, decanoic acid:tetrabutylammonium bromide, n-butylalcohol:tetrabutylammonium bromide, cyclohexanol:tetrabutylammonium bromide, 1-octanol:tetrabutylammonium bromide, 1-dodecanol:tetrabutylammonium bromide, oleyl alcohol:tetrabutylammonium bromide, hexanoic acid:methyltrioc-

Table 9. Partition Coefficients of Biomolecules between Hydrophobic DES Phase and Aqueous Phase

		Partition coefficients			
DESs	pH	Caffeine	Tryptophan	Vanillin	Isophthalic acid
DL-Menthol:Acetic acid	1.91	1.81 ± 0.01	9.05 ± 0.33	4.74 ± 0.03	8.52 ± 0.23
DL-Menthol:Lactic acid	1.18	37.33 ± 0.09	14.76 ± 0.17	3.03 ± 0.15	3.31 ± 0.12
DL-Menthol:Lactic acid	1.47	1.16 ± 0.01	2.85 ± 0.04	6.65 ± 0.04	12.94 ± 0.10
DL-Menthol:Dodecanoic acid	3.74	1.09 ± 0.01	5.23 ± 0.12	7.02 ± 0.01	1.96 ± 0.03

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tylammonium chloride, octanoic acid:methyltrioctylammonium chloride, decanoic acid:methyltrioctylammonium chloride, n-butylalcohol:methyltrioctylammonium chloride, cyclohexanol:-methyltrioctylammonium chloride, 1-octanol:methyltrioctylammonium chloride, 1-dodecanol:methyltrioctylammonium chloride, and oleyl alcohol:methyltrioctylammonium chloride. Of these combinations, the ones with cyclohexanol formed no liquid phases. It was found that the DES composed of 1-octanol:methyltrioctylammonium chloride in a 1:1 molar ratio has the highest extraction efficiency.

Hydrophobic DESs based on terpenes were used for the removal of Riboflavin, vitamin B2, from an aqueous environment.[41] Here 507 binary mixtures were tested, from which 29 liquefied at room temperature and so were considered as hydrophobic DESs. Eight of them were tested for the removal of Riboflavin from water, i.e., decanoic acid:menthol (1:1), decanoic:lidocaine (2:1), 1-tetradecanol:menthol (1:2), thymol:couramarin (2:1), thymol:coumarin (1:1), thymol:menthol (1:2), and thymol:menthol (1:1). The extraction efficiencies were compared to the one of decanoic acid:tetraoctylammonium bromide (2:1), one of the first reported hydrophobic DESs. Efficiencies varying from 20.5% to 81.1% were achieved.

Metal Ion Removal from Aqueous solutions. After the start of research into the applications of hydrophobic DESs in 2015, two papers regarding hydrophobic DESs and their applications were reported in 2016, both on the extraction of metal ions from an aqueous environment.[13,14] Tereshatov et al. reported on indium extraction with four hydrophobic DESs, i.e., tetraheptylammonium chloride:ibuprofen (7:3), tetraheptylammonium chloride:oleic acid (1:2), tetraheptylammonium chloride:decanoic acid (1:2), and DL-menthol:dodecanoic acid (2:1).[14] A kinetic study was performed to determine the time to equilibrium between the metal species in the DES phase and the water phase. It was found that this was after 5 min of shaking. The effect of a variation in HCl concentration and oxalic acid concentration on distribution coefficient was investigated. HCl concentrations varied from 0.1 to 10 M HCl; the highest distribution coefficients were achieved with 6 M of HCl in the water phase. These were 220, 280, and 600 for, respectively, tetraheptylammonium chloride:decanoic acid (1:2), tetraheptylammonium chloride:oleic acid (1:2), and tetraheptylammonium chloride:ibuprofen (7:3). Here, DL-menthol:dodecanoic acid (2:1) had a distribution coefficient lower than 0.025 (V/V). Aqueous oxalic acid concentrations varying from 1×10^{-7} to 8×10^{-1} M were tested, with the highest distribution coefficients for tetraheptylammonium chloride:ibuprofen (7:3) and tetraheptylammonium chloride:-decanoic acid (1:2) or tetraheptylammonium chloride:oleic acid (1:2), respectively, 3×10^{3} and 1.5×10^{3}. Successful back extraction to the water phase was achieved by addition of 0.1 M diethylene triamine pentaacetic acid to the water phase.

Later in 2016, van Osch et al. showed the extraction of different metal ions from an aqueous environment without the addition of acids to the water phase.[13] As hydrophobic DESs a 2:1, 3:1, and 4:1 molar ratios of decanoic acid to lidocaine were used, while the metals salts investigated were $CoCl_2$, $FeCl_2$, $MnCl_2$, $NiCl_2$, $ZnCl_2$, $Cucl_2$, NaCl, KCl, and LiCl. The results of the experiments are shown in Table 10.

Table 10 shows that divalent metal ions are always extracted with high distribution coefficients, also in the experiment with multiple metal salts present in the water phase, while only low distribution coefficients were found for the monovalent metal ions. Moreover, it was shown that an increase in the concentration of the metal salt $CoCl_2$ led to lower distribution coefficients. For the 2:1 and 3:1 ratios, this effect is minor, while for the 4:1 ratio the effect was quite large. The same trend was also observed for a change of DES/water mass ratios. It was shown that regeneration of the DES was possible, but it was determined that the mechanism of transfer occurs via an ion exchange mechanism.

Table 10. Distribution Coefficients of Metal Cations and Chloride Ions[a]

		DecA:Lid (2:1)	DecA:Lid (3:1)	DecA:Lid (4:1)
Exp. 1	Co	>0.996 ± 0.001	>0.996 ± 0.001	0.983 ± 0.002
Exp. 1	Cl	0.113 ± 0.002	0.078 ± 0.008	0.101 ± 0.059
Exp. 2	Fe	>0.992 ± 0.001	>0.991 ± 0.001	>0.991 ± 0.001
Exp. 2	Cl	0.197 ± 0.003	0.080 ± 0.001	0.113 ± 0.007
Exp. 3	Mn	>0.992 ± 0.001	>0.991 ± 0.001	0.983 ± 0.004
Exp. 3	Cl	0.086 ± 0.002	0.081 ± 0.027	0.065 ± 0.011
Exp. 4	K	0.457 ± 0.001	0.397 ± 0.011	0.457 ± 0.001
Exp. 4	Cl	0.141 ± 0.002	0.078 ± 0.031	0.072 ± 0.001
Exp. 5	Co	0.990 ± 0.001	0.946 ± 0.012	0.777 ± 0.008
Exp. 5	Ni	>0.996 ± 0.001	0.983 ± 0.001	0.880 ± 0.004
Exp. 5	Zn	>0.995 ± 0.001	>0.995 ± 0.001	>0.995 ± 0.001
Exp. 5	Cu	>0.996 ± 0.001	>0.996 ± 0.001	>0.996 ± 0.001
Exp. 5	Na	0.195 ± 0.001	0.140 ± 0.008	0.127 ± 0.040
Exp. 5	K	0.211 ± 0.028	0.161 ± 0.018	0.134 ± 0.005
Exp. 5	Li	0.226 ± 0.015	0.166 ± 0.001	0.128 ± 0.036

[a]A distribution coefficient close to zero means that hardly ions were transferred to the DES phase, while a distribution coefficient of close to one means that most of the ions are transferred to the DES phase.

Also in 2018, the removal of metals from an aqueous environment was tested with hydrophobic DESs. Hydrophobic DESs consisting of menthol/thymol and carboxylic acids with alkyl chain lengths of 8, 10, 12, 14, 16, and 18 were tested for the removal of Cu^{2+} from water.[52] For the molar fraction of decanoic acid in combination with thymol and menthol, the influence of the pH of the water phase and the concentration of Cu^{2+} were investigated in detail. From all other metal ions studied, only Fe^{2+} also showed good distribution ratios between DES and the water phase. Finally, it was shown that the solvent could be recycled.

The DES composed of decanoic acid and lidocaine was further investigated for the removal of metal ions from water in 2018.[40] It was chosen to use a 2:1 molar ratio of decanoic acid to lidocaine that was diluted with heptane. For the water phases, a 10 mmol/L of aqueous metal solution for Fe^{3+} and Mn^{2+} was prepared. It was found that a variation of the initial pH of the aqueous phase had a large influence on the removal of the metal ions. At low pHs, the mechanism of extraction was ion pair based. The DES concentration needed to remove all the Fe^{3+} is 25 g·L^{-1}, while this is 300 g·L^{-1} for Mn^{2+}, and so this solvent is very selective for the removal of Fe^{3+} over Mn^{2+}.

Another study that investigated the removal of a metal from a water phase focused on the removal of traces of the radioactive technetium in a stream containing traces of pertechnetate ($^{99m}TcO_4^-$) with three hydrophobic DESs.[35] These DESs were decanoic acid with trihexyltetradecylphosphonium chloride, decanoic acid with tetraoctylammonium bromide, and hexanoic acid with tetraoctylammonium bromide, all in a 2:1 molar ratio. To investigate the removal of the pertechnetate, different anions were added to the water phase, i.e., 0.15 M Cl^-, NO_3^-, $H_2PO_4^-$, SO_4^{2-}, I^-, or ReO_4^- at pH 5 or 0.15 M HCO_3^- at pH 8 with low concentrations of the pertechnetate. With these competing

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anions, the traces of the pertechnetate can be removed from the water phase with high partitioning efficiencies. The addition of ReO_4^- suppresses the removal of pertechnetate from the aqueous environment. Unfortunately, only limited successes were achieved for the back extraction.

For another application into the removal of metals from water, a DES was developed from phenol and trioctylphosphine oxide.[69] It was used for a liquid−liquid separation of uranyl nitrate from an aqueous environment. The removal of uranyl from water shows exceptional results with distribution ratios up to 5000 and final concentrations in the water phase lower than 0.50 ppm.

The DES composed of decanoic acid and tetrabutylammonium chloride in a 2:1 molar ratio was investigated for its chemical and electrochemical properties and used for the extraction of Cr^{6+}, Cu^{2+}, Ni^{2+}, and Cr^{3+} from an aqueous phase.[27] It was used anhydrous but also water equilibrated at the atmosphere at 30 and 60 °C. The best results for the extraction of Cr^{6+} was achieved with the water equilibrated DES at 60 °C.

Other Liquid−liquid Extractions. Menthol-based hydrophobic DESs were tested experimentally and theoretically using COSMO-SAC predictions and the nonrandom two liquid (NRTL) model for the liquid−liquid extraction of lower alcohols from aqueous solutions.[50] The DES was composed of DL-menthol and dodecanoic acid in a 2:1 ratio, while the solutes ethanol, 1-propanol, and 1-butanol were tested. With this DES, they were able to remove 5−40 mol % of ethanol, 5−30 mol % of 1-propanol, and 10−50 mol % of 1-butanol.

Hydrophobic DESs composed of thymol with ± camphor in 7:3, 3:2 and 1:1 molar ratios, thymol with 10-undecylenic acid in 7:3, 3:2, 1:1, 1:2, 1:3 and 1:4 molar ratios, and thymol with decanoic acid in 3:2, 1:1, 1:2 and 1:3 molar ratios were investigated for the extraction of polycyclic aromatic hydrocarbons from aqueous samples.[59] As an extraction method, ultrasound-assisted dispersive liquid−liquid microextraction was used. After optimization of the method, recovery rates from 73.5% to 126.2% were obtained.

The role of the hydrogen donor was investigated to ensure the development of proper hydrophobic DESs.[71] In this investigation, six hydrophobic DESs were prepared, i.e., thymol:N,N-dimethyl-N,N-didodecylammonium chloride (2:1), phenylacetic acid:N,N-dimethyl-N,Ndidodecylammonium chloride (2:1), glycolic acid:N,N-dimethyl-N,Ndidodecylammonium chloride (2:1), thymol:trimethylglycine (3:1), phenylacetic acid:trimethylglycine (2:1), and glycolic acid:trimethylglycine (2:1). These hydrophobic DESs were applied for the removal of 4-nitrophenol, 2,4-dinitrophenol, and phenol red from acidic and basic aqueous environments. The results showed that the DES composed of glycolic acid and trimethylglycine is not applicable for all tested phenolic compounds, while the one composed of glycolic acid and N,N-dimethyl-N,Ndidodecylammonium chloride (2:1) was not applicable for the removal of phenol red. For the other two phenolic compounds, the extraction efficiencies were moderate. The other hydrophobic DESs had high extraction efficiencies.

Interesting is the combination of molecular dynamics, Aspen Plus simulations, and experimental work to evaluate the removal of alcohols, i.e., 1-butanol, 1-propanol, and ethanol from water.[62] As hydrophobic DESs, dodecanoic acid:menthol in a 1:2 molar ratio and decanoic acid:menthol in a 1:1 molar ratio were tested. The highest extraction efficiencies were achieved with dodecanoic acid:menthol in a 1:2 molar ratio. With MD simulations, the extraction mechanism was investigated, which showed that menthol had the biggest influence on the extraction of the alcohols from the water phase. With Aspen Plus, a process flow sheet was constructed, which showed that after separation and recycling of the DES and alcohol, 99.7% of the DES and 98.3% of the alcohol could be recovered.

Liquid−liquid Microextraction. Liquid−liquid microextraction was used to remove synthetic pigments from beverages with hydrophobic DESs.[23] Decanoic acid:trioctylammonium chloride (2:1), octanoic acid:trioctylammonium chloride (2:1), decanoic acid:tetrabutylammonium chloride (2:1), and octanoic acid:tetrabutylammonium chloride (2:1) were used as DESs. Parameters such as type and volume of hydrophobic DES, pH, salt content, and vortex time were varied. The results showed that the DES composed of octanoic acid and tetrabutylammonium chloride performed best.

Another microextraction approach was used for the removal of nonsteroidal anti-inflammatory species from the urine samples of humans via the formation of in situ deep eutectic mixtures.[57] In detail, this method was used for measuring the amounts of diclofenac and ketoprofen in the urine of humans. The extraction recovery values were between 93% and 97%, and equilibrium was achieved within 1 min.

Hydrophobic DESs were also used for an emulsification microextraction of amphetamine and methamphetamine from human plasma and pharmaceutical wastewater.[45] As a hydrophobic DES, the combination of 1-phenylethanol and choline chloride in a 4:1 molar ratio was used. Maximum extraction efficiencies of 6%−66% were achieved with 5% of salt, 250 μL of extraction solvent, a pH of 12, and eight air agitation cycles.

The antiarrhythmic agents propranolol, carvedilol, verapamil, and amlodipine were removed from biological and environmental samples by three phase hollow fiber liquid phase microextraction.[46] The main hydrophobic DES tested was 1-phenylethanol and choline chloride in a 4:1 molar ratio. The extraction recoveries varied between 44% and 54%.

Vortex-Assisted Liquid−liquid Microextraction. Another application of hydrophobic DESs for liquid−liquid microextraction combines a vortex-assisted microextractor with an HPLC for the determination of nitrite in water and biological samples. Oleic acid and methyltrioctylammonium chloride in 1:1, 2:1, 3:1 and 1:2 molar ratios are the hydrophobic DESs used throughout this investigation.[67] As in many liquid−liquid microextraction methods, some factors are optimized to obtain the maximum recovery such as the extraction and reaction conditions. The recovery ranges as found in the investigation vary between 90.5% and 115.2%.

Liquid−liquid microextraction in combination with vortexing and analysis with the HPLC-UV approach was tested for the determination of malondialdehyde and formaldehyde, after derivatization with 2,4-dinitrophenylhydrazine, from human urine, apple juice, and rainwater.[33] Factors that influence the derivation process and the liquid microextraction were investigated with as result recoveries of 91%−112%. Formaldehyde was determined within biological and indoor air samples by hydrophobic DES-based vortex-assisted liquid−liquid microextraction with high performance liquid chromatography as the analysis method.[32] The five hydrophobic DESs investigated were hydroquinone:methyltrioctylammonium chloride in a 1:1 molar ratio, 4-phenylphenol:methyltrioctylammonium chloride in a 1:1 molar ratio, and 4-cyanophenol:methyltrioctylammonium chloride in 1:2, 1:1, and 2:1 molar ratios. The hydrophobic DES composed of 4-cyanophenol with methyltrioctylammonium chloride was used as the micro-

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extraction solvent. Under optimized conditions, 83.1%−104.4% was recovered with a standard deviation lower than 5.9%.

Dispersive Liquid−liquid Microextraction. Dispersive liquid−liquid microextraction with hydrophobic DESs controlled by freezing temperature was also investigated.[16] Basically, floating organic droplets were solidified during the dispersive liquid−liquid microextraction, after which high performance liquid chromatography was used for determining the benzoylureas in the water phase. The benzoylureas that were investigated are triflumuron, hexaflumuron, flufenoxuron, and lufenuron. Recoveries varying between 82.36% and 93.82% with high precision, relative standard deviation lower than 5% were presented.

Hydrophobic DESs were also used for the preconcentration of benzophenone-type UV filters from aqueous samples via air-assisted dispersive liquid−liquid microextraction.[22] Fifteen hydrophobic DESs were prepared, of which some components salted out at room temperature. The combinations liquid at room temperature were octanoic acid:DL-menthol (2:1), octanoic acid:DL-menthol (1:1), octanoic acid:DL-menthol (1:2), decanoic acid:DL-menthol (2:1), decanoic acid:DL-menthol (1:1), decanoic acid:DL-menthol (1:2), dodecanoic acid:DL-menthol (1:2), octanoic acid:tetrabutylammonium bromide (1:1), decanoic acid:tetrabutylammonium bromide (1:1), octanoic acid:tetrabutylammonium chloride (1:1), and decanoic acid:tetrabutylammonium chloride (1:1). It was concluded that the recovery of the DESs with ammonium salts was low, so these DESs were discarded from the investigation. From the menthol-based DESs, they found that the one composed of decanoic acid and menthol in a 1:1 ratio performed best. Factors such as the effect of the solution pH and the salt concentration were tested. They concluded that 88.8%−105.9% could be recovered with their optimized method.

Hydrophobic DESs were also used for ultrasound-assisted dispersive liquid−liquid microextraction. The extraction was used for the determination of UV filters from water samples using high-performance liquid chromatography with ultraviolet (UV) detection (HPLC-UVD).[31] The hydrophobic DESs were composed of decanoic acid and trioctylmethylammonium chloride in 1:1, 2:1, 3:1, 4:1, and 5:1 molar ratios. The analytes investigated were 2,4-dihydroxybenzophenone, benzophenone, and 2-hydroxy-4-methoxybenzophenone. They systematically investigated various factors that influenced UV filter recovery. The method designed by them showed recoveries from 90.2% to 103.5% with a good repeatability (relative standard deviation lower than 5.9%).

Another application of assisted dispersive liquid−liquid microextraction was used for the determination of the amount of Sunset Yellow and Brilliant Blue FCF present in food samples.[43] For the hydrophobic DESs, Aliquat 336 was used in combination with decanoic acid, oleic acid, and ibuprofen. The molar ratios of HBD to HBA were 2:1, 1:2, and 3:7. It was stated that this method has several advantages over conventional methods, for example, good accuracy, short extraction times, and low costs. Recoveries between 97.9% and 102.5% were found from jelly, fruit pastel, Smarties, ice cream, and candy.

Hydrophobic DESs based on hexafluoroisopropanol and l-carnitine/betaine were used as extraction solvents for the dispersive liquid−liquid microextraction of pyrethroids.[70] More specifically, the hydrophobic DESs were hexafluoroisopropanol with betaine and L-carnitine in 2:1, 2.5:1, and 3:1 molar ratios. The pyrethroids tested were transfluthrin, fenpropathrin, fenvalerate, ethofenpox, and bifenthrin, removed from tea beverages and fruit juices. The results showed that recoveries varied between 85.1% and 109.4%.

The latest work on the use of hydrophobic DESs in combination with ultrasound-assisted dispersive liquid−liquid microextraction investigated pyrethroid insecticides in environmental water.[39] The pyrethroid samples investigated were deltamethrin, fenvalerate, permethrin, etofenprox, and bifenthrin, while river water and lake water were used as the aqueous phase. For the DES phases, octanoic, decanoic, and dodecanoic acid were mixed with (tetradecyl)phosphonium tetrafluoroborate. Varying parameters, including pH value, sample volume, and salt addition were tested. The recovery ranges varied between 80.93% and 109.88%.

Microextractions Based on Solidification of Floating Drops. Since hydrophobic DESs have been described as interesting replacements for traditional solvents, research has been performed in microextractions that are based on the solidification of floating drops for the removal of polycyclic aromatic hydrocarbons from water.[65] More specifically, the carboxylic acids acetic, decanoic, acrylic, octanoic, propionic, oleic, and butyric acid were prepared in combination with tetrabutylammonium bromide in a 2:1 molar ratio of acid to the quaternary ammonium salt. The polycyclic aromatic hydrocarbons were naphthalene, fluorene, phenanthrene, anthracene, fluranthene, and pyrene. Different parameters influencing the microextraction parameters were varied, for example, extraction temperature and sample volume. It was concluded that decanoic acid:tetrabutylammonium bromide (2:1) performed the most desirable.

The microextraction research with hydrophobic DESs continued when salol:DL-menthol in a 1:1 molar ratio was researched in a microextraction method where a settled organic drop was solidified.[60] Four components were mixed with menthol in a 1:1 ratio, i.e., octanoic acid, decanoic acid, oleic acid, and salol. Salol:DL-menthol (1:1) was stated as a ultrahydrophobic DES with as properties a low viscosity and melting and freezing points and was chosen for the extraction experiments. Extractions of 2-nitrotoluene, 3-nitrotoluene, nitrobenzene, 1,3-dinitrobenzene, 1,3,5-trinitrobenzene, 2,6-dinitrotoluene, 2,4-dinitrotoluene, and 2,4,6-trinitrotoluene were investigated. After optimization of different factors, such as extraction temperature and DES composition and volume, recoveries between 89% and 105% were achieved.

The DES prepared in a 1:1 molar ratio between decanoic acid and choline chloride is used to determine the amount of lead in food and water samples by floating organic droplet microextraction.[44] The recovery rate was 95%, and the relative standard deviation was 4.2%.

Ultrasound-Assisted Emulsification Liquid Phase Microextraction. Another application in which hydrophobic DESs and microextraction were combined was the digestion and ultrasound-assisted emulsification liquid phase microextraction of copper in liver samples.[72] Seven of the eight DESs could be considered as hydrophilic, while the DES consisting of decanoic acid and tetrabutylammonium chloride in a 2:1 ratio can be considered as a hydrophobic one. Eventually, the hydrophobic DES was the most successful with a recovery percentage up to 100%.

Another application of the combination of hydrophobic DESs with ultrasound-assisted extraction was the removal of lycopene from tomato processing wastes.[73] Menthol was mixed in an 8:1 molar ratio with lactic acid to form a hydrophobic DES. An optimum yield of 1446.6 μg/g of lycopene was obtained at a

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temperature of 70 °C, 120 mL·g^{-1} of solvent over sample, and an extraction time of 10 min.

Shaker-Assisted Liquid−liquid Microextraction. Shaker-assisted liquid−liquid microextraction using hydrophobic DESs was used to remove methylene blue from alkaline solutions (0.5 M NaOH).[34] It was back-extracted into a solution consisting of hydrochloric acid (2 M). The DES used throughout the investigation was decanoic acid with methyltrioctylammonium bromide in a 2:1 molar ratio. Recovery rates between 95.91% and 102.96% were found from wastewater and river water.

Formation of a Two Phase System of Hydrophilic and Hydrophobic DESs. It was tested if a two phase system of a hydrophilic and hydrophobic DESs could be used for the extraction of bioactive compounds from *Ginkgo biloba* leaves.[56] As hydrophobic DESs, the ternary system composed of methyltrioyctylammonium chloride, capryl alcohol, and octylic acid in a 1:2:3 molar ratio was used. Three hydrophilic DESs were tested in the formation of hydrophilic−hydrophobic two phase systems, namely, levulinic acid:choline chloride (2:1) with a water content of 40%, ethylene glycol:betaine (3:1) with a water content of 40%, and malonic acid:choline chloride (2:1) with 55% of water. By mixing a hydrophobic DES with a hydrophilic DES in a 1:1 volume ratio, the two phase systems were prepared. The hydrophilic phases were single DESs or combinations of lactic acid:choline chloride (2:1) and ethylene glycol:betaine (3:1), levulinic acid:choline chloride (2:1) and malonic acid:choline chloride (2:1), or ethylene glycol:betaine (3:1) and malonic acid:choline chloride (2:1). The bioactive compounds evaluated were Ginkgo flavonoids, terpene trilactones, procyanidine, and polyprenyl acetates. For the best conditions, procyanidine (PAC), flavonoids, terpene trilactones (TTLs), and polyprenyl acetates (PPAs) were extracted at 86.07%, 77.72%, 93.29%, and 94.63%, respectively.

Two phase hydrophilic−hydrophobic DES systems were developed to investigate biphasic chiral extraction of threonine.[25] Eight hydrophobic DESs and six hydrophilic DESs were investigated. Screening of different two phase systems showed that the one of (−)-menthol-L(+)-lactic acid/choline chloride-urea gave the optimum results with a maximum enantiomeric excess value of 31.6%.

Three Phase Systems. In 2018, it was claimed that hydrophobic DESs were prepared with levulinic acid, namely, L-menthol-levulinic acid, thymol-levulinic acid, and carvacrol-levulinic acid, all in 1:1, 3:2, and a 2:3 molar ratios.[51] These were used for the preparation of ternary systems composed of the hydrophobic DES, n-heptane, and methanol. Interestingly, it was found that after equilibrium, the ratio of the HBD and HBA is not the same but changes with initial concentration of the components of the DES in the system. It is assumed that hydrogen bonding between the different components is the driving force.

Removal of Components from Leaves. In the beginning of 2017, hydrophobic DESs were used for the extraction of Artemisinin, the only effective drug in the treatment for malaria, from *Artemisia annua* leaves.[15] Thirteen DESs were used with methyl trioctylammonium chloride in combination with ethylene glycol, 1-propanol, 1,3-propanediol, glycerol, 1-butanol, 1,2-butanediol, hexyl alcohol, capryl alcohol, decyl alcohol, dodecyl alcohol, 1-tetradecanol, cyclohexanol, and DL-menthol. Since Artemisinin is a compound with moderate hydrophobicity, it was investigated what DES gave the highest extraction efficiency. Moreover, different extraction methods were investigated. The optimization steps showed that 1-butanol with methyl trioctyl ammonium chloride performed the best with a solvent/solid ratio of 17.5:1, ultrasonic power of 180 W, a temperature of 45 °C, a particle size mesh of 80, and an extraction time of 70 min as optimal extraction conditions. These optimized extraction conditions gave an extraction yield of 7.9936 ± 0.0364 mg/g and performed more efficiently than petroleum ether.

Another example in 2017 regarding the extraction from leaves was the extraction of polyprenyl acetates from *Ginkgo biloba* leaves.[30] Eighteen hydrophobic DESs were prepared, with methyl trioctylammonium chloride and hexanoic acid, octanoic acid, decanoic acid, dodecanoic acid, myristic acid, palmitic acid, cis-9-octadecenoic acid, ricinoleic acid, 1-propanol, 1-butanol, hexyl alcohol, capryl alcohol, decyl alcohol, dodecyl alcohol, 1-tetradecanol, cyclohexanol, and DL-menthol. After initial tests, the DESs with hexyl alcohol, capryl alcohol, and decyl alcohol gave the best results and were further optimized in molar ratios between the components. It was found that the hydrophobic DES composed of methyltrioctylammonium chloride, capryl alcohol, and octylic acid in a molar ratio of 1:2:3 performed the best. Several extraction methods were tested, which eventually led to an extraction yield of 84.11 ± 0.7366 mg/g.

At the start of 2018, DESs based on organic acids and quaternary ammonium salts were used in the microwave-assisted extraction of Baicalin from *Scutellaria baicalensis* Georgi.[24] The DESs used throughout this work were decanoic acid:tetrabutylammonium chloride in 2:1, 1:1, and 1:2 molar ratios, decanoic acid:methyltrioctylammonium chloride in 2:1, 1:1, and 1:2 molar ratios, decanoic acid:methyltrioctylammonium bromide in 2:1, 1:1 molar ratios, and decanoic acid:-tetraoctylammonium bromide in 2:1 molar ratios. Decanoic acid:tetrabutylammonium chloride (1:2) has the highest extraction yield with 106.96 mg·g^{-1}, which is a similar level as conducted with the pharmacopeia procedure.

Interestingly, hydrophobic DESs can also be used to remove phytocannabinoids, such as tetrahydrocannabinol, cannabidiol, and their carboxylated homologues, from raw cannabis plant material.[61] The focus was on the extraction of tetrahydrocannabinol, cannabidiol, and their carboxylated homologues. The hydrophobic DESs used throughout this investigation were produced from terpenes and natural organic acids. All the hydrophobic DESs had a 1:1 molar ratio with menthol used as HBA. The HBDs were formic acid, acetic acid, propionic acid, butyric acid, hexanoic acid, octanoic acid, dodecanoic acid, lactic acid, phenylacetic acid, and mandelic acid. The results showed that the DES composed of acetic acid and menthol performed the best, with extraction efficiencies between 118.6% and 132.6%.

Solubility of CO_2 in Hydrophobic DESs. An interesting topic addressed with decanoic acid-based hydrophobic DESs is their use as CO_2 capture agents.[68] Decanoic acid is combined with methyltrioctylammonium chloride, methyltrioctylammonium bromide, tetrabutylammonium chloride, tetraoctylammonium chloride, and tetraoctylammonium bromide in a 2:1 molar ratio. With decanoic acid and tetraoctylammonium chloride, the 1.5:1 molar ratio was tested also. The Henry's law constants for the 1.5:1 molar ratio of decanoic acid:tetraoctylammonium bromide had the highest CO_2 solubility equal to the IL $[C_4mim][BF_4]$, and it was shown that the DES could be reused for multiple cycles.

Dietz et al. modeled the solubilities of CO_2 in hydrophobic DESs with Perturbed-Chain Statistical Associating Fluid Theory (PC-SAFT).[36,78] The PC-SAFT "pseudopure" approach was

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used, and the segment number, temperature-independent segment diameter, and dispersion-energy parameter were determined based on liquid density data only. The hydrophobic DESs that they modeled were decanoic acid with tetraoctylammonium chloride in a 2:1 molar ratio, decanoic acid with tetraoctylammonium chloride in a 1.5:1 molar ratio, decanoic acid with tetraoctylammonium bromide in a 2:1 molar ratio, perfluorodecanoic acid with tetraoctylammonium chloride in a 2:1 molar ratio, and decanoic acid with tetrabutylammonium chloride in a 2:1 molar ratio. The solubilities of CO_2 in the DESs could be predicted reasonably well with a few exceptions at 298.15, 308.15, and 323.15 K.

Very recently, experimental and theoretical work has been reported on the use of hydrophobic DESs for the capture of CO_2. Ternary DESs were formed by investigation combinations between dodecanoic acid, decanoic acid, oleic acid, and tetrabutylammonium bromide. Two ternary DESs were presented: the combination of tetrabutylammonium bromide with dodecanoic acid and decanoic acid and the combination of tetrabutylammonium bromide with dodecanoic acid and decanoic acid.[75] They presented comparable CO_2 solubilities as presented before in this paragraph, but they used higher CO_2 pressures.

Interestingly, in the IL field, the nonidealities of binary mixtures of ILs have been investigated on physical absorption for the capture of CO_2.[76] It was concluded that positive deviation from ideal behavior were observed that could be used to enhance the physical absorption for CO_2 capture. This behavior should also be further investigated to see whether the solubility of DESs can be increased.

Formation of Hydrogels. Interestingly, by using a sodium DES, it was possible to form a supramolecular hydrogel.[58] A DES phase consisting of a 4:1 molar ratio of decanoic acid to dodecanoate sodium salt was used. Different regimes were found when this DES was mixed with water under variation of temperature, such as a solid, clear gel, turbid gel, two liquid phases, and a homogeneous liquid. Thus, it depends on composition and temperature what phase is formed, and it can be debated if it is truly a hydrophobic DES. It was mentioned that the physicochemical properties density and viscosity of this DES were more similar to hydrophobic than hydrophilic DESs. Finally, it was found that the viscosity is thermoreversible and depends on the water content.

Formation of Oil-in-Water Emulsions with Hydrophobic DESs. The hydrophobic DES 1-tetradecanol and menthol in a 1:2 molar ratio is investigated for the production of oil-in-water emulsions with sodium dodecyl sulfate as the surfactant.[55] It was shown that a linear correlation exists between the time and the droplet growth (d^3). Influences on the droplet growth of the surfactant concentration and the mass percentage of the DES to the water phase was investigated. The results of the hydrophobic DES as the oil phase were compared with decane as the oil phase. Interestingly, the droplet growth of the DES was lower than the one of decane, while the solubility of the DES in the water phase is many factors larger. It was shown that the DES system should not be treated as a new system but as a two component mixture and so the growth rate depends on the solubility of the less soluble component. Ideally, by using very insoluble components with a very soluble component a rather stable emulsion can be formed. So, its potential is enormous to disperse components as emulsions within an aqueous environment.

Hydrophobic DESs and Their Use in Wound Healing. Hydrophobic DESs composed of menthol in combination with lauric acid, stearic acid, and myristic acid with different molar ratios were investigated for their wound healing abilities.[74] It was shown that menthol combined with stearic acid in a 8:1 molar ratio has the lowest cytotoxicity. In general, it is concluded that menthol and stearic acid has the highest potential.

Membrane Formation. Hydrophobic DESs were used as additives for PES membranes, more specifically decanoic acid with tetrabutylammonium chloride in a 2:1 molar ratio.[26] Surprisingly, with the addition of only 2 wt % of DES, the water flux of the membrane increased to 142.84 L/(m^2·h), 4.65 times higher than the flux through the pure PES membrane. Rejection rates of pepsin, egg albumin, and bovine serum albumin were measured. They were 91.5%, 97.3%, and 99.0%, respectively.

Moreover, liquid membranes supported with a hydrophobic DES were tested for removal of furfural hydroxymethylfurfural from an aquatic environment.[18] The hydrophobic DESs tested in this work were decanoic acid:tetraoctylammonium bromide (2:1), decanoic acid:thymol (1:1), decanoic acid:menthol (1:1), and thymol:lidocaine (2:1). These hydrophobic DESs were combined with the membranes 16P10A, M3203B, and PP2E-[HF]. It was found that by addition of DES to the membrane, the transport of furfural and hydroxymethylfurfural increased.

Centrifugal Partition Chromatography. A quaternary solvent system composed of n-heptane, methanol, and the DES L-menthol with levulinic acid in a 1:1 molar ratio was used for centrifugal partition chromatography, in which it was tested whether the phases can be used as mobile and stationary phases.[63] It was found that the DES components were present in both phases. From the partition coefficient, it was concluded that the hydrophobic components were usable for centrifugal partition chromatography. Up to 79.1% of stationary phase retention was found.

Formation of a Ferrofluid. A menthol-based hydrophobic DES was combined with magnetic clay to form a ferrofluid.[48] This was coupled with directly suspended microextraction to remove explosives from water and soil samples. A long list of fatty acids in combination with DL-menthol was tested, of which the following formed a liquid phase: decanoic acid:DL-menthol (2:1), octanoic acid:DL-menthol (2:1), oleic acid:DL-menthol (2:1), dodecanoic acid:DL-menthol (2:1), valeric acid:DL-menthol (2:1), propionic acid:DL-menthol (2:1), lactic acid:DL-menthol (2:1), butyric acid:DL-menthol (2:1), acrylic acid:DL-menthol (2:1), and acetic acid:DL-menthol (2:1). The ferrofluid was made by mixing 50 mg of MMT@Fe_3O_4 magnetic nanoparticles with 1 mL of DES. Many effects were explored, such as the influence of DES type, salt addition, equilibrium time, and many more. The explosives tested were 2-nitrotoluene, 3-nitrotoluene, 1,2-dinitrobenzene, 1,3,5-trinitrobenzene, 2,6-dinitrotoluene, 2,4-dinitrotoluene, 2,4,6-trinitrotoluene, cyclotetramethylene tetranitramine, cyclotrimethylenetrinitramine, hexanitrohexaazaisowurtzitane, tetryl, and pentaerythritol tetranitrate. They found that the recoveries were between 88% and 104%.

Coatings. A hydrophobic DES was also used as an additive for sol−gel sorbent coatings of poly(dimethylsiloxane) fiber to determine the amount of toluene, ethylbenzene, and o-xylene in water.[66] Three hydrophobic DESs were prepared, i.e., 4-hydroxybenzoate:methyltrioctylammonium chloride (2:1), ethyl 4-hydroxybenzoate:methyl trioctylammonium chloride (1:1), and ethyl 4-hydroxybenzoate:methyltrioctylammonium chloride (1:2). Recovery rates of toluene, ethylbenzene, and o-

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xylene from tap, rain, well, and river water varied between 90.4% and 117.2%.

Photoluminescence and Dye Sensitized Solar Cells. In 2017, hydrophobic DESs were also used for triplet-sensitized photon upconversion.[17] By the use of photon upconversion, lower energy photons are converted into usable higher energy photons to increase solar utilization. The hydrophobic DESs decanoic acid:tetrabutylammonium chloride in 1:1, 2:1, 3:1, and 5:1 molar ratios and decanoic acid:tetraoctylammonium chloride in a 3:1 molar ratio were used. It was shown that the samples have high thermal stabilities, and a photon upconversion yield of 0.11−0.21 was found.

Three DESs, of which one was a hydrophobic DES, were used for induced circularly polarized luminescence. DESs composed of tetrabutylammonium chloride and L- and D-glutamic acid, L-proline, and L-arginine were used.[29] To measure the induced circularly polarized luminescence, a racemic mixture of dissymmetric lathanide complexes, i.e., europium, terbium, and samarium were dissolved. It was concluded that DESs, and so the hydrophobic DES, can be used as a solvent for chiral light emitting materials. By a variation in the DES components, its capability could be controlled.

The hydrophobic DES composed of acetic acid and DL-menthol was tested for its use as an electrolyte medium in the design of eco-sustainable dye sensitized solar cells.[49] It was concluded that the photovoltage was higher in comparison to commonly used volatile organic compounds. Moreover, it was suggested that an important role can be played by hydrophobic eutectic mixtures.

Catalysis. Hydrophilic and hydrophobic DESs were investigated for the production of aldols.[38] More specifically, the catalytic aldose activity was tested on 4-nitrobenzaldehyde and acetone with the porcine pancreas lipase. Four DESs were tested, i.e., glycerol:choline chloride (1.5:1), ethylene glycol:tetraoctylammonium bromide (3:1), 1,5-pentadiol:tetraoctylammonium bromide (3:1), and 4-nitrobenzaldehyde:tetrabutylammonium chloride (1.5:2.2). It was concluded that especially in hydrophobic DESs the desired aldol product was prepared. However, it was also shown that the conventional solvent acetone outperformed the DESs.

An interesting concept that was presented in 2018 where the DES has a double function acting both as solvent and product.[47] The fatty acids octanoic, decanoic, and dodecanoic acids were combined with (−)-menthol in molar ratios of 0.45:0.55, 0.35:0.65, and 0.25:0.75 at the start of the experiment. Six lipases were used, two immobilized and four solid ones. The lipases N534, c-LEcta CALB, CRL, BCL, PCL, and PFL were used. The addition of water to the reaction mixture induced the outcome of the reaction. The outcome of the addition of 10% water led to 50%, 83%, and 71% of octanoic, decanoic, and dodecanoic acids after 7 days, respectively.

CHALLENGES AND LIMITATIONS OF HYDROPHOBIC DESS

In the last 5 years, many hydrophobic DESs and various applications have been presented into the scientific literature. However, overall, it can be concluded that a major disadvantage is a lack of fundamental knowledge. It is of high importance that a proper definition of DESs be presented, which will also help to discriminate between true "deep eutectics" and merely "eutectics". First attempts have been made, but it is good if the community would come to a consensus.

It is debatable whether fundamental differences between a "deep eutectic" vs a "eutectic" makes a large difference application-wise. Although when the solid−liquid equilibria are not known, it can be tricky since solidification can occur with the smallest change in molar ratio of the DES component. Next to the solid−liquid equilibria, also the liquid−vapor equilibria should be further investigated. It is usually stated in the scientific literature that DESs are nonvolatile. However, this will depend on the components that are used and their interactions. For example, when acetic acid is used as one of the DES components, it makes common sense that the volatility is not neglectable. Regarding the liquid−liquid equilibria, these should not be taken lightly. With the addition of an aqueous phase and a solute, very complex quaternary or quinary systems are formed, which can be very challenging.

An aspect that can help to understand the fundamentals and basics of DESs is the use of theory, molecular modeling, and simulations. These aspects can strengthen the investigation of the DES field and explore the proposed factors that form a DES.

A problem related to the whole DES field is the recycling of the DESs after applications, an investigated area. This topic should be a major point of interest to ensure that DESs will have a scientific and industrial future. A type of recycling that is currently investigated is back extraction, but it is debatable if this is the best method of recycling. Most likely, new and innovative methods have to be devised to improve the recycling of the DESs.

Another point that should be addressed for the whole field of DESs is further investigation into their toxicity and biodegradability. Even when presenting DESs that are merely prepared with natural components, it is still a question if they are not toxic for the environment and the populations of the world.

A challenge truly related to hydrophobic DESs is the question whether a DES is actually hydrophobic. Many papers in the field of hydrophobic DESs use components that should not be considered as hydrophobic, for example, acetic acid and lactic acid.

A limitation related to hydrophobic DESs is the contamination of the water phase with the DES components. An important question to answer is the effect that this contamination will have on the water phase and, for example, the environment and life forms in this water phase. Most likely, the organics should be removed from the water phase. This will be a very difficult task since these concentrations are rather low.

Something interesting that is only hardly investigated is the long stability of DESs, especially between the combinations of the solid components and during prolonged use. Only recently it was investigated what the effects could be when an acid with an alcohol were mixed to form a liquid phase. It was shown that the acid and the alcohol react,[77] which basically says that it is no longer a combination of two solid components. Thus, it is not a DES anymore. Moreover, it is interesting to investigate if solid formation will occur within the solvent over a longer period of time. From an industrial aspect, a factor as corrosivity is of interest.

In general, it can be concluded that hydrophobic DESs have some unique properties and applications that make them very interesting solvents for future use in academics and industry. It is best to consider hydrophobic DESs not as the next generation of hydrophobic ILs but as a new sustainable solvent, with their own properties, applications, and challenges.

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Notes

The authors declare no competing financial interest.

Biographies



Dr. Dannie van Osch (1990) finished his M.Sc. degree in chemical engineering and chemistry in 2014 with the distinction cum laude from Eindhoven University of Technology (TU/e). In the same year, he started his Ph.D. at TU/e in the Provides project. During his 4 years as a Ph.D. candidate, he worked on the recycling of paper with hydrophobic deep eutectic solvents. His work has been published scientifically and received social attention with articles in the Dutch national news. During his academic career, Dannie has focused on sustainable chemistry and chemical engineering, which includes research into spinning disc reactors, microfluidic devices, visible light photocatalysis, sustainable solvents, and the valorization of lignin and woody biomass. In 2018, Dannie was selected as a Young European Talent. In 2019, he was chosen as a CAS Future Leader and awarded a One Young World Shell Scholarship. Since February 2019, he has been a proud member of the Brightlands Young Professionals Advisory Board. Currently, he is a postdoctoral researcher at TU/e and chief business officer (CBO) of the startup Vertoro, in which he works on the research and commercialization of crude lignin oil (CLO), the application of sustainable oil, and the valorization of lignin and lignocellulosic biomass. He is an ambassador of the Young European Talent program and One Young World. Moreover, he is an activator for IMPACTR.



Dr. Carin Dietz was born on 24-12-1968 in Venray. After finishing her degree in analytical chemistry (MLO) in 1989, she studied polymer chemistry (HLO) and graduated in 1993. Before finishing, she started working in Wageningen, industrial protein group − nonfood. In 2001, she started working at the Eindhoven University of Technology in the group of Prof. Jos Keurentjes, Process Development. In 2008, she changed to the group of Prof. Cor Koning, Polymer Chemistry. From 2012 until 2014, she has worked at Bodec in Helmond. After that, she went back to work at Eindhoven University of Technology in the Polymer and Materials Group.

Carin started her Ph.D. in 2015 under the supervision of Prof. Maaike Kroon, Prof. Martin van Sint Annaland, and Prof. Fausto Gallucci. Her Ph.D. focused on the discovery of new biobased solvents to extract FF and HMF from aqueous solutions as part of the InSciTe Horizontal project. The results obtained during this Ph.D. are presented in this dissertation. Throughout her Ph.D. time, she was involved in teaching activities. Carin also worked for one year during her Ph.D., part-time, as teacher in Applied Science at Fontys Eindhoven (2017). Now she is teacher in an analytic chemistry at Avans University.



Dr. Samah E. E. Warrag received her M.Sc. in 2014 from Texas A&M University (TAMU). Her M.Sc. thesis entitled "Multiphase Equilibrium of Fluids Confined in Fischer-Tropsch Catalytic Systems" received the Richard E. Ewing award for excellence in student research in April 2014. Upon her graduation, she joined the Eindhoven University of Technology (TUe) where she received her Ph.D. in 2018. Her Ph.D. research focused on proposing novel solutions to some industrial challenges in the purification of oil and gas utilizing low-volatility solvents from a thermodynamics point of view. The research results have been disseminated in patents and peer-reviewed scientific journals and presented at international conferences. Samah is currently a postdoctoral researcher at Khalifa University of Sciences and Technology (KUST). Her main research interests are applied thermodynamics and separation processes.

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